

FAMILIARISATION AND RELIABILITY OF AN OBJECTIVE LOWER-LIMB TESTING BATTERY FOR ANKLE REHABILITATION AND RETURN-TO-PLAY MANAGEMENT

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INTRODUCTION

More than 50% of all injuries in field- and court-based sport involve the lower limbs, with ankle ligament sprains accounting for 15% [1] and up to 40% of these progress to chronic ankle instability [2]. Despite this, there are no evidence-based criteria to inform return-to-play (RTP) decisions following lateral ankle sprains [3]. Current recommendations therefore remain predominantly time-based, with athletes typically cleared to return within 8 days [4] and a history of sprain carrying a 3.5-fold greater risk of re-injury [2].

Most objective testing used in ankle rehabilitation is adapted from anterior cruciate ligament (ACL) research. Of 114 lower extremity RTP testing studies, 82% examined ACL reconstruction and only two examined ankle injury, with isokinetic knee dynamometry most frequently reported [5]. Tests constructed to quantify quadriceps and hamstring function are unlikely to capture the plantar flexor and evtor demands central to ankle recovery.

Before ankle-specific RTP criteria can be developed, the measurement properties of the battery must be established. This study therefore quantifies (1) the number of exposures required before test performance stabilises and (2) the within- and between-session reliability of a ten-test lower-limb battery.

METHODS

A minimum of 15 recreationally active athletes aged 18 to 35 years will attend three testing sessions, each separated by seven days. Each session comprises a ten-test battery involving isometric strength tests, heel raise endurance tasks, and single-leg and bilateral repeated stationary hopping, sagittal- and frontal-plane hopping.

All tests will be measured on two in-ground force plates (Kistler, Winterthur, Switzerland; 1000 Hz) using several modules within the manufacturer's measurement, analysis, and reporting software (MARS; Winterthur, Switzerland). Testing is restricted to the preferred limb, determined by the Waterloo Footedness Questionnaire-Revised [6], to limit cumulative fatigue across a maximal battery. Training history and sleep history data are recorded as control variables.

Reliability will be assessed within session, across repeated trials, and between the three testing occasions. Relative reliability will be quantified using intraclass correlation coefficients (ICC) with 95% confidence

intervals, interpreted against a priori thresholds [7]. Absolute reliability will be expressed as coefficient of variation percentage (CV%), standard error of measurement (SEM), and minimal detectable change (MDC₉₅). Familiarisation will be evaluated between sessions 1-2 and 2-3.

RESULTS AND DISCUSSION

Data collection is ongoing; anticipated outcomes follow.

Good-to-excellent relative reliability is hypothesised for all metrics, defined as an ICC lower bound of ≥ 0.75 . Familiarisation is hypothesised to be complete by session 3, with the more novel hopping tasks requiring more exposure to remove the learning effect than the simpler bilateral jumping and isometric strength tasks.

If these hypotheses are supported, the battery will provide the SEM and MDC₉₅ values needed to distinguish genuine change in an athlete from measurement noise; the prerequisite for any objective RTP threshold to be created. Tests that fail to meet the reliability threshold will be identified for removal or refinement, and the observed familiarisation requirement will inform the number of baseline sessions required in subsequent studies.

CONCLUSIONS

Currently, the data pertaining to RTP for ankle injuries is scarce and non-validated. This ten-test battery will confirm if the objective results are reliable for future studies to develop a specific and objective RTP metric for clinicians to utilise, alongside helping to inform the testing protocol in terms of duration required to remove any learning effects.

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